

ZIYI LIANG

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RESEARCH INTERESTS

Data integration; uncertainty quantification; data science for science; multimodal learning; mobile health; computational neuroscience.

ACADEMIC POSITIONS

Postdoctoral Researcher, Department of Statistics and Applied Probability, University of California, Santa Barbara (July 12, 2026 – present)

Research advisors: Annie Qu and Babak Shahbaba

Postdoctoral Researcher, Department of Statistics, University of California, Irvine (2024 – July 11, 2026)

Research advisors: Annie Qu and Babak Shahbaba; domain-science mentor: Norbert Fortin

EDUCATION

Ph.D. in Applied Mathematics, University of Southern California (2018 – 2024)

Advisors: Matteo Sesia, Larry Goldstein

B.S. in Mathematics / Computer Science, University of California, San Diego (2014 – 2018)

Department of Mathematics

PUBLICATIONS AND PREPRINTS

*: *Equal contribution*

Work in Progress

- [1] Z. Liang, X. Cao, W. Guo, A. Qu. Conformal prediction for multi-resolution time series with application to mobile health data.
- [2] Z. Liang, A. Qu, B. Shahbaba. A Brain-inspired Reinforcement Learning Framework: Sequential Action via Graph-based Episodic Memory (SAGE).

Preprints (Under Review or Revision)

- [3] Z. Liang, H. Poursiami, Z. Yang, K. W. Cooper, A. Jaiswal, M. Parsa, N. Fortin, B. Shahbaba. ConformalHDC: Uncertainty-Aware Hyperdimensional Computing with Application to Neural Decoding. *Preprint*. <https://arxiv.org/abs/2602.21446>
- [4] Z. Liang, A. Qu, B. Shahbaba. Meta Fusion: A Unified Framework For Multimodality Fusion with Mutual Learning. *Preprint*. <https://arxiv.org/abs/2507.20089>
- [5] T. Xie, Y. Zhou, Z. Liang, S. Favaro, M. Sesia. Conformal Inference for Open-Set and Imbalanced Classification. *Preprint*. <https://arxiv.org/abs/2510.13037>

Journal Articles

- [6] Z. Liang*, T. Xie*, X. Tong, M. Sesia. Structured Conformal Inference for Matrix Completion with Applications to Group Recommender Systems. *Journal of the American Statistical Association* (published online, 2026). <https://doi.org/10.1080/01621459.2026.2658287>

- [7] Z. Liang, T. T. Cai, W. Sun and Y. Xia. A Locally Adaptive Algorithm for Multiple Testing with Network Structure. *Statistica Sinica* 36 (2026). <https://doi.org/10.5705/ss.202024.0002>
- [8] Z. Liang, M. Sesia, W. Sun. Integrative Conformal Inference for Out-of-Distribution Testing with Labeled Outliers. *Journal of the Royal Statistical Society: Series B* (2024). <https://doi.org/10.1093/jrsssb/qkad138>

Conference and Workshop Papers

- [9] Z. Moslemi, Z. Liang, N. Fortin, B. Shahbaba. Multi-Graph Meta-Transformer: An Interpretable Framework for Cross-Graph Functional Alignment in Neural Decoding. *Advances in Neural Information Processing Systems (NeurIPS)* (2025). Paper
- [10] Z. Liang*, Y. Zhou*, M. Sesia. Conformal inference is (almost) free for neural networks trained with early stopping. *International Conference on Machine Learning (ICML)* (2023). Paper
- [11] S. Bucci*, J. I. D. Centeno*, B. Gaffinet*, Z. Liang*, V. Bickel, B. Moseley, M. Olivares-Mendez. Thermophysical Change Detection on the Moon with the Lunar Reconnaissance Orbiter Diviner sensor. *NeurIPS 2022 Workshop on Machine Learning and the Physical Sciences* (2022). https://ml4physicalsciences.github.io/2022/files/NeurIPS_ML4PS_2022_174.pdf

HONORS AND AWARDS

Theodore Edward Harris Teaching Prize (2024).

Awarded by Department of Mathematics, University of Southern California.

OTHER PROFESSIONAL ACTIVITIES

Machine Learning Research Intern, Frontier Development Lab, San Jose, California. (Summer 2022)

TEACHING EXPERIENCE

University of Southern California:

Instructor: MATH 040 — Basic Mathematical Skills (Fall 2023).

Teaching Assistant: MATH 541A — Introduction to Mathematical Statistics (Spring 2023).

Teaching Assistant: MATH 114 — Foundations of Statistics (Fall 2020, Fall 2021).

Teaching Assistant: MATH 408 — Mathematical Statistics (Spring 2021).

Teaching Assistant: MATH 129 — Calculus II for Engineers and Scientists (Spring 2019).

SELECTED PRESENTATIONS

Conference and Workshop Presentations

ICML 2023 (Poster). Conformal inference is (almost) free for neural networks trained with early stopping. (July 2023)

NeurIPS 2022 (Poster). Thermophysical Change Detection on the Lunar Surface. (Dec 2022)

Seminar Presentations

USC Probability Seminar. Transfer learning for multiple testing with applications. (March 2022)

USC Math Department. Review on dimensionality reduction techniques and applications of random projections. (Dec 2019)

Industry Presentations

FDL 2022 Showcase. Change Detection on the Moon with the LRO Diviner sensor. (Aug 2022)

PROFESSIONAL SERVICE

Referee for the *Journal of the American Statistical Association*.